**Zadanie 2.** $\mathbf{E} = \hat{x}\,3\cos(\omega t - kz)$ **[V/m]. Korzystając z równań Maxwella wyznaczyć wektor H.**

Próżnia: $\mathbf{D} = \varepsilon_0\mathbf{E}$, $\mathbf{J} = \sigma\mathbf{E} = 0$. Prawo Ampère'a z poprawką Maxwella i prawo Gaussa dla $\mathbf{B}$:

$$\nabla \times \mathbf{H} = \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \qquad \nabla \cdot \mathbf{H} = 0$$

Lewa strona (rotacja jako wyznacznik):

$$\nabla \times \mathbf{H} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = \hat{x}\left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right) + \hat{y}\left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x}\right) + \hat{z}\left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right)$$

Prawa strona: $\varepsilon_0 \dfrac{\partial \mathbf{E}}{\partial t} = \hat{x}\,\varepsilon_0 \dfrac{\partial E_x}{\partial t} = -\hat{x}\,3\varepsilon_0\omega\sin(\omega t - kz) \qquad (E_y = E_z = 0)$

**Równania składowe:**

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = -3\varepsilon_0\omega\sin(\omega t - kz)$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = 0$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = 0$$

$$\frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z} = 0$$

Fala biegnie w kierunku $z$, więc $\dfrac{\partial}{\partial x} = \dfrac{\partial}{\partial y} = 0$, $\dfrac{\partial}{\partial z} \neq 0$. Stąd:

$$-\frac{\partial H_y}{\partial z} = -3\varepsilon_0\omega\sin(\omega t - kz)$$

$$\frac{\partial H_x}{\partial z} = 0 \Rightarrow H_x = 0, \qquad 0 = 0, \qquad \frac{\partial H_z}{\partial z} = 0 \Rightarrow H_z = 0$$

Całkowanie po $z$:

$$H_y = 3\varepsilon_0\omega \int \sin(\omega t - kz)\,dz = \frac{3\varepsilon_0\omega}{k}\cos(\omega t - kz)$$

**Odp.** $\mathbf{H} = \hat{y}\,\dfrac{3\varepsilon_0\omega}{k}\cos(\omega t - kz)$ [A/m]

$\dfrac{\varepsilon_0\omega}{k} = \varepsilon_0 c = \dfrac{\varepsilon_0}{\sqrt{\mu_0\varepsilon_0}} = \sqrt{\dfrac{\varepsilon_0}{\mu_0}} = \dfrac{1}{Z_0}$: **Odp.** $\mathbf{H} = \hat{y}\,\dfrac{3}{Z_0}\cos(\omega t - kz)$ [A/m]

**Zadanie 3.** $\mathbf{H} = \hat{y}\,3\sin(\omega t + kx)$ **[A/m]. Korzystając z równań Maxwella wyznaczyć wektor E.**

Prawo Faradaya: $\nabla \times \mathbf{E} = -\mu_0 \dfrac{\partial \mathbf{H}}{\partial t}, \qquad \nabla \cdot \mathbf{E} = 0$

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \hat{x}\left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right) + \hat{y}\left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right) + \hat{z}\left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right)$$

Prawa strona: $-\mu_0 \dfrac{\partial \mathbf{H}}{\partial t} = -\hat{y}\,3\omega\mu_0 \cos(\omega t + kx)$

**Równania składowe:**

$$\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = 0$$

$$\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = -3\omega\mu_0 \cos(\omega t + kx)$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = 0$$

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0$$

Fala biegnie w kierunku $-x$, więc $\dfrac{\partial}{\partial y} = \dfrac{\partial}{\partial z} = 0,\ \dfrac{\partial}{\partial x} \neq 0.$ Stąd:

$$-\frac{\partial E_z}{\partial x} = -3\omega\mu_0 \cos(\omega t + kx), \qquad \frac{\partial E_y}{\partial x} = 0 \Rightarrow E_y = 0, \qquad \frac{\partial E_x}{\partial x} = 0 \Rightarrow E_x = 0$$

Całkowanie po $x$:

$$E_z = 3\mu_0\omega \int \cos(\omega t + kx)\,dx = \frac{3\mu_0\omega}{k} \sin(\omega t + kx)$$

$$\frac{\mu_0\omega}{k} = \mu_0 c = \frac{\mu_0}{\sqrt{\mu_0\varepsilon_0}} = \sqrt{\frac{\mu_0}{\varepsilon_0}} = Z_0$$

**Odp.** $\mathbf{E} = \hat{z}\,3Z_0 \sin(\omega t + kx)$ [V/m]

Kontrola: $|\mathbf{E}|/|\mathbf{H}| = Z_0,\ \ \hat{z} \times \hat{y} = -\hat{x}$ — zgodne z kierunkiem ruchu fali ($-x$).

**Rozwiązanie wykonane niezależnie; zgadza się z wynikiem w zeszycie.**